

Subspace-Based Adaptive Detection under Spatially Correlated Heavy-Tailed Sea Clutter

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Abstract—This article studies the problem of subspace-based adaptive target detection in spatially correlated heavy-tailed sea clutter. The clutter texture is modeled using an inverse gamma distribution to capture the heavy-tailed characteristics of sea clutter. For the case of completely correlated texture, a subspace-based generalized likelihood ratio test (CIG-SGLRT) is derived by assuming full range correlation. For partially correlated scenarios, spatial correlation in the inverse gamma texture is introduced via a memoryless nonlinear transformation (MNL), leading to the proposed MNIG-SGLRT detector. An iterative extension based on a coordinate descent scheme is further developed to exploit the correlation among secondary cells. Simulation results demonstrate that the proposed detectors provide improved detection performance compared with conventional methods.

Index Terms—Spatially-correlated sea clutter, inverse gamma texture, subspace detection, memoryless nonlinear transformation (MNL).

I. INTRODUCTION

Adaptive radar target detection in sea clutter is a long-standing problem in maritime radar signal processing due to its importance in surveillance and navigation applications [1], [2]. With the improvement of radar resolution and operating conditions, sea clutter returns increasingly exhibit heavy-tailed and non-Gaussian behavior, which can severely degrade the performance of conventional Gaussian-based detectors [3], [4]. This has motivated extensive research on statistical clutter modeling and adaptive detection methods.

Among various statistical models, the compound Gaussian (CG) model has been widely adopted to characterize sea clutter by representing it as the product of a slowly varying non-Gaussian texture and a fast-varying complex Gaussian speckle component [1]. Early CG models typically assume gamma-distributed texture, leading to the well-known K-distribution [5]. However, high-resolution sea clutter measurements have revealed more pronounced heavy-tailed characteristics, making the K-distribution inadequate in many practical scenarios [2]. As a result, alternative texture models, such as the inverse gamma (IG) distribution, have been proposed to better capture the statistical properties of sea clutter [6].

Based on CG clutter models with IG texture, a variety of adaptive detectors have been developed within the generalized likelihood ratio test (GLRT) framework [6], [7]. Nevertheless, most existing works assume that the texture samples are independent and identically distributed across range cells. In practice, experimental studies have demonstrated that sea clutter

texture often exhibits significant spatial correlation along the range dimension [2], [8], and neglecting this correlation may lead to performance degradation.

To model texture spatial correlation in non-Gaussian clutter, several approaches have been proposed. Simplified models assume identical texture within a local region, resulting in a completely correlated texture structure that is analytically tractable but overly restrictive [9]. More flexible techniques, such as the memoryless nonlinear transformation (MNL), introduce spatial correlation by transforming a correlated Gaussian process into a non-Gaussian texture while preserving the desired marginal distribution [10], [11]. The MNL approach has been shown to effectively reproduce realistic spatial correlation patterns observed in measured sea clutter.

Motivated by these observations, this article investigates adaptive target detection in compound Gaussian clutter with spatially correlated inverse gamma texture. GLRT-based detectors are first developed for the completely correlated texture case. To address more general scenarios involving partially correlated textures, the MNL framework is naturally incorporated into the GLRT formulation to model spatial texture correlation, while subspace signal representations are adopted to accommodate targets with multiple degrees of freedom. The main contributions of this work are summarized as follows:

- GLRT-based adaptive subspace detectors are developed for CG clutter with IG texture by explicitly accounting for texture spatial correlation, including both completely correlated models and partially correlated models realized through the integration of the MNL method.
- Simulation results demonstrate that incorporating texture spatial correlation yields notable performance improvements over conventional detectors that assume independent texture samples.

In this paper, we denote scalars by regular lowercase or uppercase letters (e.g., a , A), vectors by bold lowercase letters (e.g., $\mathbf{a}$), and matrices by bold uppercase letters (e.g., $\mathbf{A}$). The superscripts $(\cdot)^T$, $(\cdot)^H$, and $(\cdot)^{-1}$ denote the transpose, Hermitian transpose, and matrix inverse, respectively. The identity matrix of size $N \times N$ is denoted by $\mathbf{I}_N$. The operator $\text{tr}(\cdot)$ denotes the matrix trace, and $\det(\cdot)$ denotes the determinant. The notation $\mathbb{E}[\cdot]$ denotes the statistical expectation, and $|\cdot|$ denotes the modulus (absolute value) of a real or complex scalar. The Kronecker product is denoted by $\otimes$. The

complex field is denoted by $\mathbb{C}$, and $\mathbb{C}^{M \times N}$ denotes the space of all $M \times N$ complex-valued matrices. $\mathcal{CN}(\boldsymbol{\mu}, \mathbf{R})$ denotes a complex Gaussian distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\mathbf{R}$.

II. PROBLEM FORMULATION

Consider a radar system transmitting N coherent pulses. Target detection in CG sea clutter is formulated as the binary hypothesis test

$$\begin{aligned} H_0 : & \begin{cases} \mathbf{z} = \mathbf{c} \\ \mathbf{z}_k = \mathbf{c}_k, \quad k = 1, \dots, K \end{cases} \\ H_1 : & \begin{cases} \mathbf{z} = \mathbf{H}\boldsymbol{\theta} + \mathbf{c} \\ \mathbf{z}_k = \mathbf{c}_k, \quad k = 1, \dots, K \end{cases} \end{aligned} \quad (1)$$

where $\mathbf{z} \in \mathbb{C}^N$ is the primary data vector from the cell under test (CUT), and $\mathbf{z}_k \in \mathbb{C}^N$, $k = 1, \dots, K$, denote the secondary data vectors from neighboring range cells.

The clutter components are modeled using the CG framework as

$$\mathbf{c} = \sqrt{\tau} \mathbf{g}, \quad \mathbf{c}_k = \sqrt{\tau_k} \mathbf{g}_k,$$

where τ and τ_k are nonnegative texture variables, and $\mathbf{g}, \mathbf{g}_k \sim \mathcal{CN}(\mathbf{0}, \mathbf{R})$ represent the speckle components with covariance matrix $\mathbf{R}$. The target signal is modeled as $\mathbf{H}\boldsymbol{\theta}$, where $\mathbf{H} \in \mathbb{C}^{N \times p}$ is a full-column-rank steering matrix and $\boldsymbol{\theta}$ is the unknown target coefficient vector.

The texture variable τ is assumed to follow an inverse gamma (IG) distribution with PDF

$$f_\tau(\tau) = \frac{\beta^\alpha}{\Gamma(\alpha)} \tau^{-\alpha-1} \exp\left(-\frac{\beta}{\tau}\right), \quad (2)$$

where α and β denote the shape and scale parameters, respectively.

In practical sea clutter environments, texture samples across range cells are often spatially correlated. Let $\tau(k)$ denote the texture in the k th range cell. The normalized spatial correlation coefficient is defined as

$$R_\tau(l) = \frac{\mathbb{E}[\tau(k)\tau(k+l)] - |\mathbb{E}[\tau(k)]|^2}{\mathbb{E}[\tau(k)^2] - |\mathbb{E}[\tau(k)]|^2}, \quad (3)$$

which quantifies the dependence between texture samples separated by l range cells. Ignoring this correlation may result in significant performance degradation, motivating the explicit incorporation of texture correlation into detector design. When $R_\tau(l)$ decays from 1 to $1/e$, the texture components are typically considered to be no longer correlated [2], [9], [12].

III. DETECTOR DESIGN

We first introduce a completely correlated texture model and derive a corresponding IG texture subspace GLRT detector. For partially correlated textures, the MNLT method is employed to model spatial correlation, based on which adaptive and iterative subspace detectors are developed.

A. Completely Correlated Subspace Detector

Under hypotheses H_0 and H_1 , the received data $\mathbf{z}$ follow complex Gaussian distributions conditioned on the texture variable τ , with covariance matrix $\tau\mathbf{R}$ and mean $\mathbf{0}$ or $\mathbf{H}\boldsymbol{\theta}$, respectively. Assuming $\mathbf{R}$ is known (or estimated from secondary data), the GLRT reduces to

$$\Phi_{\text{CIG-SGLRT}} = \frac{\max_{\boldsymbol{\theta}} f(\mathbf{z}|\tau, \boldsymbol{\theta}; H_1)}{f(\mathbf{z}|\tau; H_0)}. \quad (4)$$

Maximization with respect to $\boldsymbol{\theta}$ yields the standard subspace projection

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^H \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^H \mathbf{R}^{-1} \mathbf{z}, \quad (5)$$

leading to the test statistic

$$\Phi_{\text{CIG-SGLRT}} = \frac{1}{\tau} \mathbf{z}^H \mathbf{R}^{-1} \mathbf{H} (\mathbf{H}^H \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^H \mathbf{R}^{-1} \mathbf{z}. \quad (6)$$

To exploit texture information from neighboring range cells, we assume the texture component is completely correlated across the CUT and L secondary cells, and L is obtained when the correlation function $R_\tau(L/2)$ in (3) drops to $1/e$. The joint likelihood under H_0 then depends on the quadratic forms $\mathbf{z}^H \mathbf{R}^{-1} \mathbf{z}$ and $\mathbf{z}_i^H \mathbf{R}^{-1} \mathbf{z}_i$, $i = 1, \dots, L$.

Assuming an IG prior for τ , its MAP estimate is given by

$$\hat{\tau} = \frac{\mathbf{z}^H \mathbf{R}^{-1} \mathbf{z} + \sum_{i=1}^L \mathbf{z}_i^H \mathbf{R}^{-1} \mathbf{z}_i + \beta}{N(L+1) + \alpha + 1}. \quad (7)$$

Substituting (7) into (6) and discarding constant terms, the final CIG-SGLRT statistic becomes

$$\Phi_{\text{CIG-SGLRT}} = \frac{\mathbf{z}^H \mathbf{R}^{-1} \mathbf{H} (\mathbf{H}^H \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^H \mathbf{R}^{-1} \mathbf{z}}{\mathbf{z}^H \mathbf{R}^{-1} \mathbf{z} + \sum_{i=1}^L \mathbf{z}_i^H \mathbf{R}^{-1} \mathbf{z}_i + \beta}. \quad (8)$$

B. Partially Correlated Subspace Detector

The MNLT method generates a non-Gaussian sequence by mapping a correlated Gaussian process, preserving correlation and the desired marginal distribution. The transformation is defined as

$$\begin{aligned} \tau &= \mathcal{T}(x) = Q_{\text{dist}}\left(\text{erfc}\left(x/\sqrt{2}\right)/2\right), \\ x &= \mathcal{T}^{-1}(\tau) = \sqrt{2} \text{erfc}^{-1}[2 - 2F_\tau(\tau)], \end{aligned} \quad (9)$$

where F_τ is the cumulative distribution function (CDF) of the IG distribution [13], x is the zero-mean unit-variance correlated Gaussian process, $Q_{\text{dist}}(\zeta)$ is the complementary quantile function defined as

$$Q_{\text{dist}}(\zeta) = F_\tau^{-1}(1 - \zeta), \quad (10)$$

and $\text{erfc}(x)$ is the complementary error function defined as

$$\text{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty \exp(-t^2) dt. \quad (11)$$

Using (10) and (11), the correlation mapping between the IG process and the corresponding correlated Gaussian process can also be computed. This mapping is explicitly represented as follows [14]

$$\mathbb{E}(\tau(k)\tau(k+l)) = \frac{1}{\pi} \sum_{n=0}^{+\infty} \frac{R_x(l)^n}{2^n n!} I_n^2, \quad (12)$$

where $R_x(l)$ denotes the ACF of the Gaussian process, and I_n is defined as

$$I_n = \int_{-\infty}^{+\infty} \exp(-x^2) H_n(x) Q_{dist}(\text{erfc}(x)/2) dx, \quad (13)$$

with H_n denoting the Hermite polynomial of order n as

$$H_n(x) = (-1)^n \exp(x^2) \left(\frac{d}{dx} \right)^n \exp(-x^2) \quad (14)$$

In practice, the infinite series in (13) is truncated to a finite order, which has been verified to provide sufficiently accurate correlation matching in our simulations.

By applying the MNLT method, the partially correlated inverse-gamma texture vector $\boldsymbol{\tau} = [\tau, \tau_1, \dots, \tau_K]^T$ is mapped to an underlying correlated Gaussian vector $\mathbf{x} = [x, x_1, \dots, x_K]^T$ via the inverse transformation $x = \mathcal{T}^{-1}(\tau)$ and $x_i = \mathcal{T}^{-1}(\tau_i)$. The joint PDF of $\boldsymbol{\tau}$ is therefore given by

$$f(\boldsymbol{\tau}) = f_x(\mathbf{x}) \frac{dx}{d\tau} \prod_{i=1}^K \frac{dx_i}{d\tau_i}, \quad (15)$$

where $f_x(\mathbf{x})$ denotes the joint PDF of the corresponding Gaussian process,

$$f_x(\mathbf{x}) = \frac{1}{(2\pi)^{(K+1)/2} [\det(\mathbf{R}_x)]^{1/2}} \exp\left(-\frac{1}{2} \mathbf{x}^T \mathbf{R}_x^{-1} \mathbf{x}\right), \quad (16)$$

with covariance matrix $\mathbf{R}_x$ constructed as a toeplitz matrix from the autocorrelation function in (12) to capture the spatial correlation among texture components.

The MAP estimation of textures under H_0 is

$$\hat{\boldsymbol{\tau}} = \arg \max_{\boldsymbol{\tau}} f(\mathbf{z}, \mathbf{z}_1, \dots, \mathbf{z}_K | \boldsymbol{\tau}; H_0) f(\boldsymbol{\tau}). \quad (17)$$

To reduce the computational complexity of the MAP estimation, we first fix the secondary texture components $[\tau_1, \dots, \tau_K]$ to their individual MAP estimates

$$\hat{\tau}_i = \frac{\mathbf{z}_i^H \mathbf{R}^{-1} \mathbf{z}_i + \beta}{N + \alpha + 1}, \quad i = 1, \dots, K, \quad (18)$$

and then, the MAP estimation of τ using the MNLT model can be written as

$$\hat{\tau}_M = \arg \max_{\tau} f(\mathbf{z}, \mathbf{z}_1, \dots, \mathbf{z}_K | \tau, \hat{\tau}_1, \dots, \hat{\tau}_K; H_0) f(x | \hat{x}_1, \dots, \hat{x}_K) \frac{dx}{d\tau}, \quad (19)$$

where $\hat{x}_i = \mathcal{T}^{-1}(\hat{\tau}_i)$, and $f(x | \hat{x}_1, \dots, \hat{x}_K)$ follows a conditional Gaussian distribution $N(\mu_x, \sigma_x^2)$ with $\mu_x = \mathbf{r}_{12} \mathbf{R}_{22}^{-1} \hat{\mathbf{x}}$ and $\sigma_x^2 = r_{11} - \mathbf{r}_{12} \mathbf{R}_{22}^{-1} \mathbf{r}_{21}$, where

$$\begin{bmatrix} r_{11} & \mathbf{r}_{12} \\ \mathbf{r}_{21} & \mathbf{R}_{22} \end{bmatrix} = \mathbf{R}_x, \quad \hat{\mathbf{x}} = [\mathcal{T}^{-1}(\hat{\tau}_1), \dots, \mathcal{T}^{-1}(\hat{\tau}_K)]^T. \quad (20)$$

In (20), r_{11} denotes the (1, 1) element of the $\mathbf{R}_x$, $\mathbf{r}_{12} \in \mathbb{R}^{1 \times K}$, $\mathbf{r}_{21} = \mathbf{r}_{12}^T$, and $\mathbf{R}_{22} \in \mathbb{R}^{K \times K}$. Taking the natural logarithm of the right side of (19) and ignoring the irrelevant terms unrelated to τ , we can obtain

$$\eta(\tau) = -\frac{\mathbf{z}^H \mathbf{R}^{-1} \mathbf{z} + \beta}{\tau} - \frac{[\mathcal{T}^{-1}(\tau) - \mu_x]^2}{2\sigma_x^2} - (N + \alpha + 1) \ln(\tau) + \frac{\mathcal{T}^{-1}(\tau)^2}{2}. \quad (21)$$

Nulling the derivative of τ in (21), we have

$$\left. \frac{d\eta}{d\tau} \right|_{\tau=\hat{\tau}_M} = \frac{\mathbf{z}^H \mathbf{R}^{-1} \mathbf{z} + \beta}{\hat{\tau}_M^2} - \frac{N + \alpha + 1}{\hat{\tau}_M} + \phi(\hat{\tau}_M) = 0, \quad (22)$$

where $\phi(\hat{\tau}_M)$ is represented as

$$\phi(\hat{\tau}_M) = \left[\mathcal{T}^{-1}(\hat{\tau}_M) - \frac{(\mathcal{T}^{-1}(\hat{\tau}_M) - \mu_x)}{\sigma_x^2} \right] \left. \frac{dx}{d\tau} \right|_{\tau=\hat{\tau}_M}. \quad (23)$$

Several numerical or iterative methods [15], [16] can be used to solve (22). Then, the texture estimate $\hat{\tau}_M$ for the partially correlated case can be obtained, which is subsequently substituted into (6) to compute the MNIG-SGLRT test statistic

$$\Phi_{MNIG-SGLRT} = \frac{\mathbf{z}^H \mathbf{R}^{-1} \mathbf{H} (\mathbf{H}^H \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^H \mathbf{R}^{-1} \mathbf{z}}{\hat{\tau}_M}. \quad (24)$$

The effectiveness of the MNLT-based approach is strongly influenced by the precision of texture estimation from the reference cells. To improve this estimation, we propose a coordinate descent algorithm that exploits the spatial correlation among the textures of reference cells. By iteratively refining each texture estimate, this method significantly improves the detection performance of the MNLT-based spatial correlation detector. The iterative procedure is outlined as follows

$$\begin{cases} \hat{\tau}^{(i)} &= \arg \max_{\tau} f(\tau, \hat{\tau}_1^{(i-1)}, \dots, \hat{\tau}_K^{(i-1)} | \mathbf{z}, \mathbf{z}_1, \dots, \mathbf{z}_K; H_0) \\ \hat{\tau}_1^{(i)} &= \arg \max_{\tau_1} f(\hat{\tau}^{(i)}, \tau_1, \hat{\tau}_2^{(i-1)}, \dots, \hat{\tau}_K^{(i-1)} | \mathbf{z}, \mathbf{z}_1, \dots, \mathbf{z}_K; H_0) \\ \dots & \\ \hat{\tau}_K^{(i)} &= \arg \max_{\tau_K} f(\hat{\tau}^{(i)}, \hat{\tau}_1^{(i)}, \dots, \hat{\tau}_{K-1}^{(i)}, \tau_K | \mathbf{z}, \mathbf{z}_1, \dots, \mathbf{z}_K; H_0), \end{cases} \quad (25)$$

where $\hat{\tau}_k^{(i)}$ is the estimation of τ_k at the i th iteration. The iteration is initialized using the closed-form MAP estimates obtained under the independence assumption in (18), and continues until convergence or a maximum number of iterations is reached. The resulting iterative MAP estimate of the CUT texture is denoted by $\hat{\tau}_{IM}$, which is then substituted into (6), the IT-MNIG-SGLRT test statistic is finally obtained as

$$\Phi_{IT-MNIG-SGLRT} = \frac{\mathbf{z}^H \mathbf{R}^{-1} \mathbf{H} (\mathbf{H}^H \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^H \mathbf{R}^{-1} \mathbf{z}}{\hat{\tau}_{IM}}. \quad (26)$$

In the simulation experiment, for generality, we randomly generate $\mathbf{H}$ and $\boldsymbol{\theta}_0$. Then, we set $\boldsymbol{\theta} = \alpha \boldsymbol{\theta}_0$, where α satisfies

$$\alpha = \sqrt{\frac{\mathbb{E}[\tau] \text{SCR}}{\boldsymbol{\theta}_0^H \mathbf{H}^H \mathbf{R}^{-1} \mathbf{H} \boldsymbol{\theta}_0}}. \quad (27)$$

In this article, we use approximate maximum likelihood (AML) estimator [17] to estimate the speckle covariance matrix $\mathbf{R}$, which uses an iterative method as follows

$$\boldsymbol{\Sigma}_{AML}(i+1) = \frac{N}{K} \sum_{k=1}^K \frac{\mathbf{z}_k \mathbf{z}_k^H}{\mathbf{z}_k^H \hat{\mathbf{R}}^{-1}(i) \mathbf{z}_k}, \quad (28)$$

$$\hat{\mathbf{R}}(i+1) = N \frac{\boldsymbol{\Sigma}_{AML}(i+1)}{\text{tr}(\boldsymbol{\Sigma}_{AML}(i+1))}. \quad (29)$$

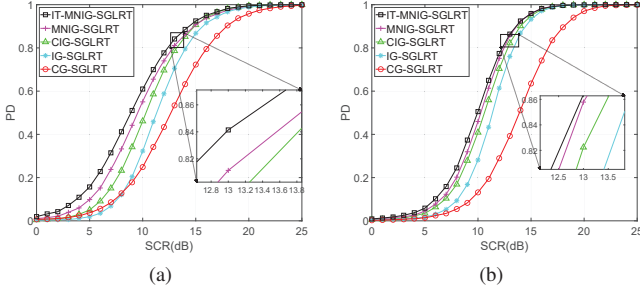


Fig. 1: PDs versus SCR with simulated data at different shape parameters. $\text{PFA} = 10^{-3}$, $N = 8$, $K = 32$. (a) $\alpha = 2.5$; (b) $\alpha = 5$.

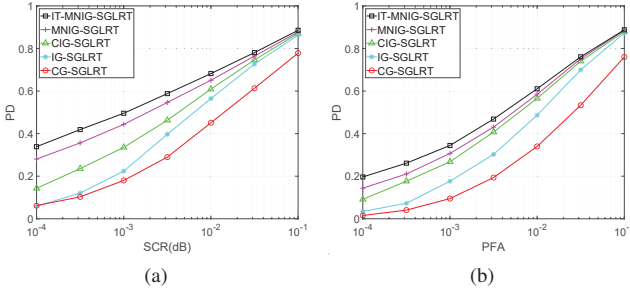


Fig. 2: ROC curves with simulated data at different shape parameters. $N = 8$, $\text{SCR} = 9$ dB, $K = 32$. (a) $\alpha = 2.5$; (b) $\alpha = 5$.

The initial value of $\hat{\mathbf{R}}$ is set to the normalized sample covariance matrix (NSCM), which is written as

$$\hat{\mathbf{R}}(0) = \frac{N}{K} \sum_{k=1}^K \frac{\mathbf{z}_k \mathbf{z}_k^H}{\mathbf{z}_k^H \mathbf{z}_k}. \quad (30)$$

IV. SIMULATION EXPERIMENTS

In this section, we evaluate the detection performance of the proposed CIG-SGLRT, MNIG-SGLRT, and IT-MNIG-SGLRT detectors using simulated data. For comparison, we also consider the IG-SGLRT [18] and CG-SGLRT [19] detectors, which are subspace-based adaptive detectors designed for point target detection in compound Gaussian clutter with IG texture without spatial correlation. The detection threshold and the probability of detection (PD) are estimated via Monte Carlo simulations with 10^4 independent trials for PD and 100/PFA trials for threshold estimation. The texture correlation function follows an exponential model $R_\tau(l) = v^{|l|}$ [20], where v is set to 0.8 unless otherwise specified. The coherent processing interval (CPI) length is set to $N = 8$ and the clutter speckle covariance matrix is modeled as $\mathbf{R}(m, n) = \rho^{|m-n|}$, where ρ denotes the one-lag correlation coefficient [17], and m, n are the pulse indices. β only affects the overall amplitude of the clutter, it is fixed to $\beta = 1$ without loss of generality.

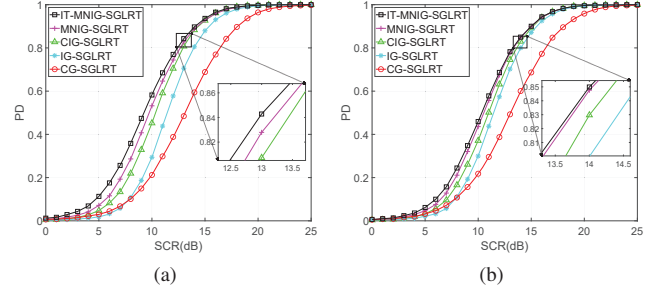


Fig. 3: PDs versus SCR with simulated data at different correlation coefficients. $\alpha = 3$, $\text{PFA} = 10^{-3}$, $N = 8$, $K = 32$. (a) $v = 0.9$; (b) $v = 0.6$.

The signal-to-clutter ratio (SCR) is defined as [6]

$$\text{SCR} = \frac{\boldsymbol{\theta}^H \mathbf{H}^H \mathbf{R}^{-1} \mathbf{H} \boldsymbol{\theta}}{\mathbb{E}[\tau]}, \quad \mathbb{E}[\tau] = \begin{cases} \frac{\beta}{\alpha-1}, & \alpha > 1 \\ \beta, & \alpha = 1. \end{cases} \quad (31)$$

Fig. 1 shows the PD versus SCR for different shape parameters $\alpha = 2.5$ and $\alpha = 5$ with $K = 32$ and $\text{PFA} = 10^{-3}$. Fig. 2 presents the corresponding receiver operating characteristic (ROC) curves at a fixed SCR of 9 dB, which clearly highlight the performance differences among different detectors. Both figures show that the proposed spatially correlated detectors consistently outperform the IG-SGLRT and CG-SGLRT. This performance advantage is more pronounced for smaller α , demonstrating that exploiting texture spatial correlation is particularly effective in heavy-tailed clutter. In addition, the results in Fig. 1 further indicate that the MNLT-based detectors demonstrate superior performance compared to the CIG-SGLRT detector in low SCR region, indicating that the MNLT-based approach effectively captures the range correlation characteristics of sea clutter.

Fig. 3 illustrates the PD versus SCR of the proposed detectors under different texture spatial correlations modeled as $R_\tau(l) = v^{|l|}$, with $\alpha = 3$ and $\text{PFA} = 10^{-3}$. Two correlation scenarios are considered: strong correlation $v = 0.9$ in Fig. 3(a) and weak correlation $v = 0.6$ in Fig. 3(b). The results show that the performance gain of the proposed detectors over correlation-agnostic baselines increases with stronger texture correlation, while the advantage of the iterative MNLT-based approach diminishes in low-correlation scenarios.

V. CONCLUSION

This paper addressed adaptive point target detection in spatially correlated compound Gaussian sea clutter. A GLRT-based detector was first developed for the completely correlated texture case. The MNLT framework was then employed to model spatially correlated clutter texture and derive an adaptive subspace detector, with a coordinate descent scheme introduced to improve texture estimation accuracy. Simulation results demonstrate the effectiveness of the proposed detectors over conventional methods.

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